

| Call for Expressions of Interest                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <i>Date of Issue:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | September 2026                                                                                |
| <i>Expected Completion Date:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | May 2027                                                                                      |
| <i>Project Title:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Real-Time AI Sign-Language Avatar                                                             |
| <i>Company:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | UoN                                                                                           |
| Contact Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                               |
| <i>Lead Contact:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Shreyank N Gowda                                                                              |
| <i>Lead E-mail:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Shreyank.narayanagowda@nottingham.ac.uk                                                       |
| <i>Additional Contact:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Additional contact name                                                                       |
| <i>E-mail:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Additional contact e-mail                                                                     |
| <i>Additional Contact:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Additional contact name                                                                       |
| <i>E-mail:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Additional contact e-mail                                                                     |
| <i>Additional Contact:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Additional contact name                                                                       |
| <i>E-mail:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Additional contact e-mail                                                                     |
| Additional Info                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                               |
| <i>Students must sign NDA:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <input type="checkbox"/>                                                                      |
| <i>Project uses personal data:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <input type="checkbox"/>                                                                      |
| <i>Custom hardware required:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <input type="checkbox"/><br>If Applicable: Please provide details                             |
| <i>Relevant research areas (if applicable)<sup>1</sup>:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Artificial Intelligence, NLP, HCI, Computer Graphics, Web/Software Engineering, Accessibility |
| Project Background Information                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                               |
| <p>Sign languages are rich visual languages that use coordinated hand shape, movement, body posture and facial expression. Providing sign-language access at scale is difficult, and many digital services still rely on text, captions or human interpretation. Recent progress in language models, computer graphics and real-time 3D animation creates an opportunity to investigate how software can turn written or spoken input into understandable signing performed by a virtual avatar.</p> <p>The project will develop a web-based prototype that accepts text and produces a sequence of signs rendered by a 3D avatar. The initial focus should be British Sign Language (BSL) and a deliberately limited vocabulary so that the team can concentrate on good software engineering, integration and user experience rather than attempting a full translation system. The system should be modular: an input and language-processing layer maps text to an intermediate sign representation; an animation layer selects and blends sign motions; and a web application presents the result interactively. A simple backend should manage sign assets, configuration, logging and performance metrics.</p> <p>The technical challenge is not a single AI model, but the design and integration of several components into a reliable end-to-end application. Students can</p> |                                                                                               |

---

<sup>1</sup> An overview of different research areas in the school of computer science can be found [here](#).

work across frontend and backend development, NLP, 3D animation, data pipelines, testing, accessibility and deployment. Existing open datasets, models and animation assets may be used where appropriately licensed. The core project should not require students to collect new signer data or use personal data.

The project is intentionally extensible. A strong team could add speech input, improved linguistic processing, facial expressions, motion blending, alternative avatars, a larger vocabulary, analytics, evaluation tools or support for another sign language. This gives a clear minimum viable product while leaving substantial scope for an ambitious team to create a polished and impressive demonstration.

## Requirements

*The group must provide:*

- A functional web application that accepts text and renders a corresponding sequence of BSL signs using a real-time or near-real-time 3D avatar.
- A modular architecture separating the user interface, language/sign-sequence processing, animation/rendering and data/storage components.
- A demonstrable sign vocabulary of approximately 300-500 signs, subject to the availability and quality of suitable licensed data/assets.
- Smooth hand and upper-body animation, with basic non-manual features (for example facial expression or head movement) where feasible.
- A sign/animation asset-management pipeline so that new signs can be added without redesigning the whole system.
- Basic performance and usage measurements, such as processing/rendering latency and successful/failed sign lookups.
- Automated and manual testing of the major software components, with technical documentation and deployment instructions.
- A final interactive demonstration showing the complete text-to-avatar pipeline.

Potential features / extensions:

- Speech-to-text input and live microphone mode.
- More advanced NLP for mapping English input into a sign-language-friendly representation.
- Motion blending, inverse kinematics, improved facial animation or alternative avatar styles.
- Playback controls, camera controls and accessibility preferences.
- A larger vocabulary, domain-specific phrase packs, or experimental support for another sign language.
- An evaluation interface for anonymous comprehension/naturalness feedback, subject to appropriate approval if human participants are involved.

## Skills

*Highly desirable:*

*Frontend and backend development; software architecture; web APIs; JavaScript/TypeScript or Python; interest in AI/NLP; 3D*

*Desirable:*

*Project management; UX/accessibility; databases; cloud deployment; basic machine learning; user testing; technical documentation; interest in sign language or accessible technology.*

|                                                                                               |                |
|-----------------------------------------------------------------------------------------------|----------------|
| <i>graphics/animation (e.g. Three.js, Unity or similar);<br/>version control and testing.</i> |                |
| <b>Submission / Award information</b>                                                         |                |
| <i>Date of Submission:</i>                                                                    | September 2026 |
| <i>Date of Pitch:</i>                                                                         | October 2026   |
| <i>Notification of award:</i>                                                                 | October 2026   |

For further information on this proposal, the team representative should in the first instance e-mail the lead contact stated above at the e-mail address provided.